

States of Matter – Criterion A

Q1. The diagrams below show three substances.

- A: particles are closely packed and arranged regularly.
- B: particles are close together but can move past one another.
- C: particles are far apart and move freely.

Select the substance that represents a liquid.

- A. Substance A
- B. Substance B
- C. Substance C

Answer: B. Substance B

In a **liquid**, particles are **close together** but are able to **move/slide past one another**.

Scientific terminology: liquid • particles • close together • move past one another

Q2. Recall your scientific knowledge about liquids. Describe the arrangement, spacing and movement of particles in a liquid.

Answer:

In a liquid, the particles are **close together** with **small spaces** between them. They have a **random arrangement** and are not held in fixed positions. The particles can **move and slide past one another**, which allows a liquid to **flow** and take the shape of its container.

Scientific terminology: particles • close together • small spaces • random arrangement • move past one another • flow

Q3. State why gases can be compressed easily but liquids cannot be compressed easily. Give a scientific reason using the spacing between particles.

Answer:

Gases can be **compressed** easily because gas particles are **far apart** and have **large spaces** between them. When pressure is applied, the particles can be pushed closer together.

Liquids cannot be compressed easily because their particles are already **close together**, with **very little space** between them.

Scientific terminology: compressed • particles • large spaces • far apart • close together • pressure

Q4. Water vapour in the air changes directly into tiny ice crystals on the inside wall of a freezer. Using the particle model, outline what happens to the movement, arrangement and energy of the particles during this change. Name the change of state.

Answer:

The change of state from **gas directly to solid** is called **deposition**.

As water vapour cools, its particles **lose thermal energy** and therefore have less **kinetic energy**. The particles move more slowly and come **closer together**.

They change from being **far apart and randomly arranged** in the gas to being **closely packed in a fixed, regular arrangement** in the solid ice crystals. In the solid state, the particles no longer move freely; they **vibrate about fixed positions**.

Scientific terminology: deposition • gas → solid • thermal energy • kinetic energy • fixed arrangement • vibrate about fixed positions

Q5. A student wants to prevent an ice cube from melting quickly during a science activity.

She can place it:

Option A: on a metal tray near a sunny window

Option B: inside a closed insulated container

Select the better option and suggest why it would help the ice cube remain solid for longer.

Answer:

Option B – inside a closed insulated container is better.

The insulated container **reduces the transfer of thermal energy** from the warmer surroundings to the ice cube. Therefore, the ice cube warms more slowly and takes longer to **melt**, allowing it to remain solid for longer.

Scientific terminology: insulation • thermal energy • heat transfer • melting • solid

Q6. A family wants to freeze homemade juice into ice pops as quickly as possible.

They can place the mould:

A: near the freezer door, where the temperature changes frequently

B: deep inside the freezer, where the temperature remains very low

Apply your knowledge of freezing to suggest the better position. Give a scientific reason.

Answer:

Position B – deep inside the freezer is better.

The temperature is **lower and more constant**, so the juice loses **thermal energy** more quickly. As the particles lose energy, their **kinetic energy decreases**, so they move more slowly and eventually become held in **fixed positions**.

The liquid therefore changes into a solid by **freezing** more quickly.

Scientific terminology: freezing • thermal energy • kinetic energy • fixed positions • liquid → solid

Q7. A shopkeeper notices that frozen food begins to soften when the freezer door is left open for a long time.

Apply your knowledge of particle theory to explain why the frozen food begins to melt and solve the problem by suggesting one action the shopkeeper should take to reduce melting.

Answer:

When the freezer door is left open, warmer air enters the freezer and **thermal energy is transferred** to the frozen food.

The particles in the frozen food **gain kinetic energy** and vibrate more rapidly. When enough energy is absorbed, the particles overcome some of the **forces of attraction** holding them in fixed positions and begin to move past one another. The solid therefore starts **melting into a liquid**.

The shopkeeper should **keep the freezer door closed as much as possible**. This reduces the transfer of thermal energy from the warmer surroundings and helps keep the food below its **melting point**.

Scientific terminology: thermal energy • kinetic energy • forces of attraction • melting • melting point • heat transfer

Q8. On very cold nights, water vapour in the air can change directly into ice crystals and form frost on young plants. Frost can damage the plants.

a) Identify the change of state that forms frost and describe the change in particle movement, arrangement and energy.

Answer:

The change of state is **deposition**, where a **gas changes directly into a solid**.

The water vapour particles **lose thermal energy** and their **kinetic energy decreases**. They move more slowly and come closer together. The particles change from being **far**

apart and randomly arranged to being **closely packed in a fixed arrangement**. In the solid ice crystals, the particles **vibrate about fixed positions**.

b) Suggest one method the farmer could use to protect the plants from frost and explain scientifically how the method would help.

Answer:

The farmer could **cover the plants with frost cloth, fabric or another insulating cover**.

The covering reduces **heat transfer** from the plants to the colder surroundings and traps a layer of warmer air around them. This helps keep the temperature around the plants higher and reduces the formation of frost.

Scientific terminology: deposition • gas → solid • thermal energy • kinetic energy • fixed arrangement • insulation • heat transfer

Q9. A student observes three substances.

Substance	Observation
A	Keeps its own shape
B	Takes the shape of its container but keeps the same volume
C	Spreads out to fill the entire container

A student says:

“Substance B is a gas.”

Apply the information in the table to make a judgment about whether the statement is correct. Give one piece of evidence from the table.

Answer:

The student's statement is **incorrect**.

Substance B is a **liquid**, not a gas. The evidence is that it **takes the shape of its container but keeps the same volume**.

A gas would spread out to **fill the entire container** and would not have a fixed volume.

Scientific terminology: liquid • gas • fixed volume • shape of container

Q10. Two students leave equal amounts of ice in different places.

Cup	Location	Time taken for ice to melt
A	Sunny windowsill	12 minutes
B	Shaded table	28 minutes

A student concludes:

"Ice melts faster when it receives more thermal energy."

Apply the information in the table to make a judgment about the student's conclusion. Give a scientific reason for your judgment.

Answer:

The student's conclusion is **supported by the evidence**.

The ice on the **sunny windowsill melted in 12 minutes**, while the ice on the **shaded table took 28 minutes**.

The ice on the sunny windowsill received more **thermal energy** from the surroundings. Its particles gained more **kinetic energy** and vibrated more rapidly. This caused the ice to reach its **melting point** sooner and change from a **solid to a liquid**.

Therefore, the results support the conclusion that greater thermal energy can increase the rate of **melting**.

Scientific terminology: thermal energy • kinetic energy • melting • melting point • solid → liquid

Q11. A student places equal amounts of water in three containers.

Container	Surface area	Water remaining after 3 hours
A	Small	45 mL
B	Medium	38 mL
C	Large	29 mL

All three containers started with **50 mL of water** and were kept at the same temperature.

The student concludes:

"Increasing the surface area increases the rate of evaporation."

Apply the information to make a scientifically supported judgment about the student's conclusion. In your answer, use two pieces of evidence from the table and use particle theory.

Answer:

The student's conclusion is **supported by the evidence**.

Container A, with the **small surface area**, had **45 mL** remaining, meaning only **5 mL evaporated**.

Container C, with the **large surface area**, had only **29 mL** remaining, meaning **21 mL evaporated**.

Therefore, more water evaporated from the container with the larger surface area.

According to **particle theory**, evaporation occurs when particles at the **surface of a liquid** have enough **kinetic energy** to overcome the **forces of attraction** and escape into the air as **water vapour**.

A larger surface area exposes **more water particles to the air at the same time**, so more particles can escape. Therefore, increasing surface area increases the **rate of evaporation**.

Scientific terminology: evaporation • surface area • kinetic energy • surface particles • forces of attraction • water vapour • rate of evaporation

Q12. A student investigates how temperature affects the melting of equal-sized ice cubes.

Trial	Temperature of surroundings	Size of ice cube	Time taken to melt
A	15°C	Equal	22 min
B	25°C	Equal	14 min
C	35°C	Equal	7 min

The student concludes:

“Increasing the temperature of the surroundings makes ice melt faster.”

Interpret the information in the table to make a scientifically supported judgment about the student's conclusion.

In your answer:

1. Use at least two pieces of evidence from the table.
2. Use particle theory to explain the pattern.
3. Decide whether the evidence is sufficient to support the student's conclusion.

Answer:

The evidence **supports the student's conclusion** that increasing the temperature of the surroundings makes ice melt faster.

At **15°C**, the ice cube took **22 minutes** to melt. When the temperature increased to **25°C**, the melting time decreased to **14 minutes**. At the highest temperature of **35°C**, the ice cube melted in only **7 minutes**. This shows a clear pattern: **as temperature increases, melting time decreases**.

According to **particle theory**, at a higher surrounding temperature, **thermal energy is transferred** to the ice more rapidly. The particles in the ice gain **kinetic energy** and vibrate more rapidly. When sufficient energy is absorbed, the particles overcome some of the **forces of attraction** holding them in fixed positions. The solid then changes into a liquid by **melting**.

The evidence is **sufficient to support the conclusion for the conditions tested** because the ice cubes were **equal in size**, while the temperature was changed systematically from **15°C → 25°C → 35°C**, and the melting time consistently decreased from **22 → 14 → 7 minutes**. However, repeating each trial would make the evidence **more reliable**. MYP -1 Integrated Science-FA-1...

Scientific terminology: thermal energy • kinetic energy • particle theory • forces of attraction • fixed positions • melting • melting point • evidence • reliability • controlled variable